

TEA-CART: How-to Guide

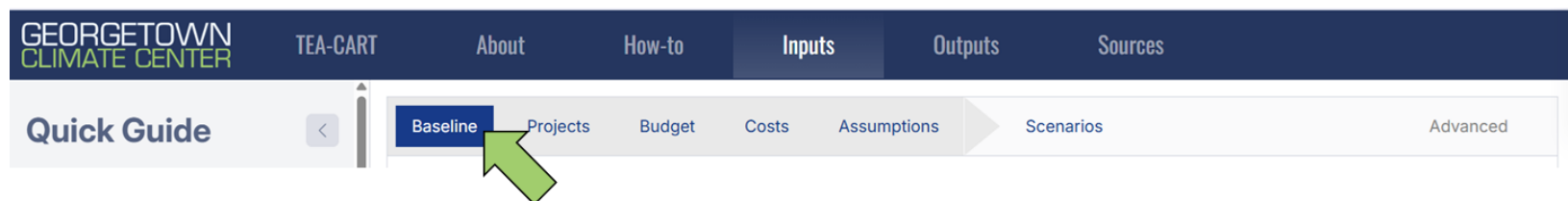
Guidance on using the [tool](#)

As you enter data, please keep the following in mind:

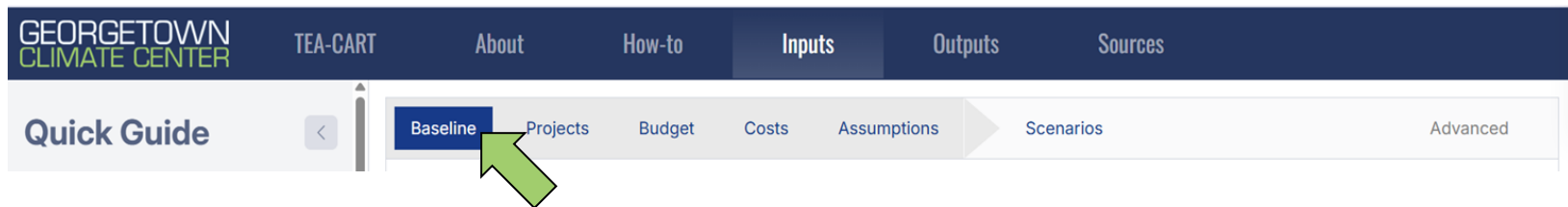
- This application does not require a login and is free to use.
- In the left side panel, there is a button to **Download User Inputs**. *It is essential that you download data prior to long periods of inactivity*. If you need to leave the application before completing the analysis, you can later upload the saved file to continue your work.
- Data are entered in tables. To begin entering (or editing) data, double click anywhere in the table using your mouse. When you are done entering data, press **CTRL + ENTER** on your keyboard (this initiates the calculation). Data can only be entered for one table at a time.
- When you are ready to see the results of your analysis, proceed to the **Scenarios** tab and select scenarios for analysis. Either choose groups of projects (to support a comparison) or click the button to select all projects for each scenario.

Steps to use the tool

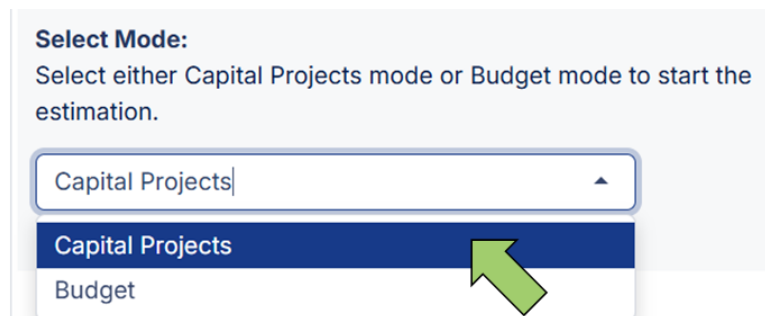
1. Use the navigation panel at the top to select **Inputs**.



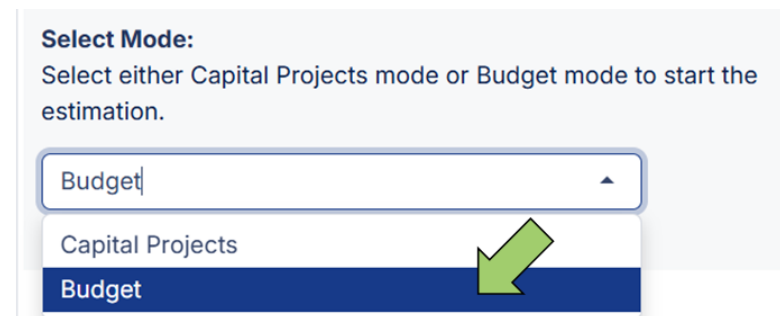
2. Within **Inputs**, select **Baseline** to choose your state and enter the years used for the planning forecast. You may also change some parameters for the forecast, the scope of emissions to include (for example, whether to include certain upstream emissions), and other assumptions. Advanced users may choose to edit the downloaded data file directly, as an alternative method for generating user inputs to upload.



3. At the bottom of the **Baseline** tab, if you would like to provide project-level inputs, click on **Capital Projects** under the **Select Mode** dropdown, and proceed to *step 4*. Alternatively, if you would like to provide budget-level inputs, click on **Budget** under the dropdown, and skip to *step 6* below.



OR



4. Within **Projects**, enter information about each project. To begin entering (or editing) data, double click in the table using your mouse.

Projects 1 | Bicycle & Pedestrian Lane Miles of New Infrastructure
^

Reset Projects 1

This category represents implementation of any **two-way miles of new bicycle or pedestrian facility**. The default assumption for these project types is that any new bicycle or pedestrian facility would be two-way. For one-way facilities, please enter half the total miles for the facility.

Search:

Area Type	Facility Type	Unit	2025	2030	2050
Core	New shared-use path	New lane-miles	0	0	0
Urban	New shared-use path	New lane-miles	0	0	0
Suburban	New shared-use path	New lane-miles	0	0	0
Rural	New shared-use path	New lane-miles	0	0	0
Core	New bicycle lane	New lane-miles	0		0

Double-click the value (here, a '0') to begin editing.

- When you are done entering data, press **CTRL + ENTER** on your keyboard (note, your cursor must be in a cell, inside the table, when you do this). Data can only be entered for one table at a time.

Projects 1 | Bicycle & Pedestrian Lane Miles of New Infrastructure

Reset Projects 1

This category represents implementation of any **two-way miles of new bicycle or pedestrian facility**. The default assumption for these project types is that any new bicycle or pedestrian facility would be two-way. For one-way facilities, please enter half the total miles for the facility.

Search:

Area Type	Facility Type	Unit	2025	2030	2050
Core	New shared-use path	New lane-miles	0	0	0
Urban	New shared-use path	New lane-miles	0	0	0
Suburban	New shared-use path	New lane-miles	10	20	0
Rural	New shared-use path	New lane-miles			0
Core	New bicycle lane	New lane-miles			0

Hit **Ctrl + Enter** to finish entering, or **Esc** to cancel.

6. In the **Budget** tab, data can be entered the same way as in the **Projects** tab. If you already entered data in the **Projects** tab, go to *step 8* below. If, instead, you would like to provide budget-level inputs, double click with your mouse in the Budget tables to begin entering values. Once you are done, press **CTRL + ENTER** on your keyboard (note, again, your cursor must be in a cell, inside the table, when you do this). As a reminder, data can only be entered one table at a time.

Budget 1 | Bicycle & Pedestrian Lane Miles of New Infrastructure ⓘ

This category represents spending on any **two-way miles of new bicycle or pedestrian facility**, expressed as a percentage (%) of the total budget (shown at the top of this tab). The default assumption for these project types is that any new bicycle or pedestrian facility would be two-way. For one-way facilities, please enter half the budget amount.

Reset Budget 1

Search:

Area Type	Facility Type	Value
Core	New shared-use path	0
Urban	New shared-use path	0
Suburban	New shared-use path	0.25
Rural	New shared-use path	0
Core	New bicycle lane	0

Similarly, in Budget mode, **double-click** cells to begin editing.

7. After you are done entering **Budget** values, you must click on **Fill Projects Tab with Budget Inputs** (at the top of the **Budget** tab), to initiate the calculation.

Baseline

Projects

Budget

Costs

Assumptions

Scenarios

Advanced

Please provide **budget-level inputs** for one or more of the project categories shown below. Make sure that **Budget** is selected under **Select Mode**, in the **Baseline** tab.

Alternatively, go to the Projects tab if you would like to provide *project-level* inputs instead.

Outputs are not generated automatically. After filling out the budget **click on the Fill Project Tab with Budget Inputs button in order to generate outputs**.

Keep in mind the following two rules when entering budget-level inputs:

- i) All spending is assumed to be equally distributed over the years of spending by the corresponding horizon year.
- ii) Spending estimates are assumed to be averages based on historical data.

Funding Start Year: ⓘ

Total Years Covered: ⓘ

Total Budget (\$M) for 2026 - 2031:

2026

5

100

Fill Projects Tab with Budget Inputs

8. Regularly download the data that you have entered. You can download it by clicking the **Download User Inputs** button on the sidebar.

Quick Guide

This sidebar has a high-level step-by-step guide for entering and saving data in TEA-CART.

New User

- 1 Navigate to the **Inputs** tab.
- 2 Input your values in the **Baseline, Projects / Budget, Costs, and Assumptions** tabs.
- 3 When you are done entering data, press **CTRL + Enter** on your keyboard to initiate the calculations.
- 4 Select the desired combination of strategies in the **Scenarios** tab.
- 5 Navigate to the **Outputs** tab to view your results.

Returning User

Upload User Inputs

Note: Set tool to Budget mode if entering a budget scenario.

Save My Work

9. If desired, within **Costs**, enter custom unit costs for the project type. Note that default values have already been provided.

Baseline

Projects

Budget

Costs

Assumptions

Scenarios

Advanced

This tab provides information on the **cost inputs** for the project categories shown below.

All costs are already populated with default values. **Users do not need to edit this tab in order for the tool to work.**

To edit cost information, please click on the different fields to overwrite the default values with any custom values provided by the user.

Note that there is no **Costs 4** or **Costs 17**. Costs related to transit electrification can be found in **Costs 2** (with other transit vehicle costs). Costs related to transit service cuts cannot be updated by the user.

All dollar values are in **2024 dollars**.

Costs 1 | Bicycle & Pedestrian Costs

Reset Costs 1

This category represents the **overall cost per mile** of bicycle or pedestrian facilities being implemented.

Search:

Area Type	Facility Type	Cost per Facility Mile	Annual Maintenance Cost per Facility Mile
Core	New shared-use path	566955	56695.5
Urban	New shared-use path	566955	56695.5
Suburban	New shared-use path	566955	56695.5

Default values
can be changed.

10. Further **Assumptions** for the analysis are for advanced users. Data can be changed similar to the previous tabs. For more information about changing assumptions, please refer to the User Guide and Methodology Documentation.

Baseline Projects Budget Costs **Assumptions** Scenarios Advanced

This tab provides information on the **input assumptions** for the categories shown below. These assumptions affect the GHG impact and effectiveness of each strategy category. To better understand which assumptions relate to which strategies, refer to the User Guide & Methodology Documentation.

All assumptions are already populated with default values. **Users do not need to edit this tab in order for the tool to work.**

To edit assumptions, please click on the different fields to overwrite the default values with any custom values provided by the user.

Assumptions 1 | Bicycle & Pedestrian Parameters ^

Reset Assumptions 1

This category includes the following data types:

- (i) **Prior drive mode share of new walkers / bikers**, which represents the fraction of new walkers / bikers who would have previously driven a car. This is calculated as a percentage of the new walkers / bikers or "persons per square mile" (ppsm) within the different area types defined.
- (ii) **Average trip length**, which represents the average Walk and / or Bike trip length in miles.
- (iii) **Annualization**, which represents the annual number of days multiplier for the mode-shift to walking / biking.

Element	Unit	Value
Annualization	Annualization factor	365
Walk	Average trip length	0.7
Bike	Average trip length	2.2

Default values can be changed.

11. In the **Scenarios** tab, click the boxes to choose which groups of projects to include in the scenario analysis.

Baseline Projects Budget Costs Assumptions **Scenarios** Advanced

This tab enables the user to select different combinations of strategies – based on *Project* or *Budget* inputs – to include in the results displayed on the Outputs tabs.

The user can select up to two “scenarios” with different combinations of project or budget inputs.

Scenario Selections

Search:

	Grouped Projects	Relation to Project Tab	Scenario1	Scenario2
1	Bicycle and Pedestrian	Projects 1	☒	☐
2	Transit Service Expansion	Projects 2, Projects 3, Projects 5, Projects 6	☒	☐
3	Transit Electrification	Projects 4	☒	☒
4	Travel Demand Management	Project	☒	☒
5	Micromobility	Project	☒	☐

By default, all project types will be selected under Scenario 1. Click the checkboxes to add or remove project types.

12. It is possible to use a custom forecast on the **Advanced** tab for the number of EVs on the road (for example, for states that have a goal to add one million EVs to the road by a certain deadline), future VMT, and other advanced parameters. Refer to the User Guide and Methodology Documentation for more information.

13. To see the results of the data that have been entered, refer to the **Outputs** tab.



14. Click **Download Output Tables** in the sidebar to get an Excel file with all of this information.

Quick Guide

This sidebar has a high-level step-by-step guide for entering and saving data in TEA-CART.

New User

- 1 Navigate to the **Inputs** tab.
- 2 Input your values in the **Baseline, Projects / Budget, Costs, and Assumptions** tabs.
- 3 When you are done entering data, press **CTRL + Enter** on your keyboard to initiate the calculations.
- 4 Select the desired combination of strategies in the **Scenarios** tab.
- 5 Navigate to the **Outputs** tab to view your results.

Returning User

Upload User Inputs

Note: Set tool to Budget mode if entering a budget scenario.

Save My Work